

Streamflow Prediction in Ungauged Basins: Review of Regionalization Methods

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Abstract: The paper presents a comprehensive review of a fundamental and challenging issue in hydrology: the regionalization of streamflow and its advances over the last two decades, specifically 1990-2011. This includes a discussion of developments in continuous streamflow regionalization, model parameter optimization methods, the application of uncertainty analysis in regionalization procedures, limitations and challenges, and future research directions. Here, regionalization refers to a process of transferring hydrological information from gauged to ungauged or poorly gauged basins to estimate the streamflow. Huge efforts have been devoted to regionalization of flood peaks, low flow and flow duration curves (FDCs) in the literature, while continuous streamflow regionalization is helpful in deriving each of these variables. Continuous streamflow regionalization can be conducted through rainfall-runoff models or hydrologic model-independent methods. In the former case, model parameters are used as instruments to transfer hydrological information from gauged to ungauged basins; while the latter case transfers streamflow directly through data-driven methods.

According to the reviewed regionalization studies, streamflow regionalization has been done mostly through hydrologic models, while the focus of these studies is on identifying the best methods to transfer the model parameters. Conceptual rainfall-runoff models such as HBV (Hydrologiska Byråns Vattenbalansavdelning) and IHACRES (Identification of unit Hydrographs And Component flows from Rainfall, Evaporation and Streamflow data) have emerged as the most frequently used models in this category. Physiographic attributes (e.g. catchment area, elevation, slope of basins or channels) and meteorological information (e.g. daily time series of rainfall and temperature) are the most commonly used in the regionalization studies. Diversity in catchment physical attributes and climatic variability

produces different performances for each regionalization method's application in various regions. However, overall, spatial proximity and physical similarity have shown satisfactory performance in arid to warm temperate climate (e.g. Australia) and regression-based methods have been preferred in warm temperate regions (e.g. most European countries). Similarly, in cold and snowy regions (e.g. Canada) spatial proximity and physical similarity approaches seemed to be good options among the hydrologic model-dependent methods. Hydrologic model-independent methods have been applied only in few cases and the results have indicated that in warm temperate regions linear and nonlinear regression methods perform well.

Author keywords: Continuous streamflow, Regionalization, Ungauged basins, Rainfall-runoff models, Uncertainty analysis, Parameter optimization

Introduction

Continuous streamflow estimation is an important issue in surface hydrology, especially in ungauged watersheds. According to Sivapalan et al. (2003), ungauged basins are ones with inadequate records (in terms of both data quantity and quality) of hydrological observations. A catchment is ungauged or poorly gauged with respect to a variable of interest. The International Association of Hydrological Science (IAHS) initiated the decade 2003-2012 with the Prediction in Ungauged Basins (PUB), defined as the prediction or forecasting of the hydrological responses of ungauged or poorly gauged basins and its associated uncertainty (Sivapalan et al.

2003). Most of the studies in this context were conducted after 2003. An example of the earlier studies is Burn and Boorman's (1993) work that investigated hydrologic model parameters estimation based on the classification of catchments according to hydrologic similarity, rather than traditionally used methods such as regression equations. Uncertainty analysis in PUB has also gone relatively less addressed.

The majority of rivers and stream reaches and tributaries in the world are ungauged or poorly gauged (Sivapalan et al. 2003; Young 2006; Mishra and Coulibaly 2009). Such ungauged streams are often located upstream (e.g. mountain areas in Niadas 2005), in maritime and mountainous Apenninic regions (e.g. Castellarin et al. 2007), in 'unregulated' basins (e.g. Stainton and Metcalfe 2007; Mishra and Coulibaly 2009), or in rural and remote areas (e.g. Makungo et al. 2010). Although ungauged basins are often located upstream due to inaccessibility or lack of developer intentions; this state also occurs in many potential sites downstream (Goswami et al. 2007).

Reliable continuous streamflow forecasting is an important factor in watershed planning and sustainable water resource management, as it is instrumental in obtaining a deeper sense of flow variability in ungauged basins. Furthermore, the estimation of flood peaks, low flow characteristics, and FDCs can be derived once the synthetic continuous flow time series is generated. Streamflow data is also used in the design of critical engineering structures such as highways, drainage systems, reservoirs, etc.

Streamflow in gauged and/or ungauged basins is currently forecasted using distributed physically-based models (e.g. BTOPMC [Block wise use of TOPMODEL with Muskingum-Cunge flow routing method], Mike 11 NAM, MIKE-SHE , etc.), conceptual and semi-distributed models (e.g. HBV, SimHyd, IHACRES), and data-driven models (e.g. MLR

[Multiple Linear Regression], ARMA [Auto Regressive Moving Average], ANNs [Artificial Neural Networks]). When predictions of streamflow response are required, less complex conceptual lumped models have been shown to be equally reliable and are often preferred (Yadav et al. 2007). The application of physically-based models in which model parameters are derived from physical catchment attributes to ungauged catchments is associated with high levels of uncertainty, reflecting uncertainty in the prior distribution of parameter values (Bulygina et al. 2011). This sometimes causes over-parameterization and model structural errors (Goswami et al. 2007; Yadav et al. 2007). Such models require considerable data and human effort compared to conceptual or semi-distributed models, such as the HBV model (Bergström 1976) and SAC-SMA (Sacramento Soil Moisture Accounting, Burnash et al. 1973). Therefore, usually non-distributed models are preferred. One example of applying physically-based models for streamflow simulation in ungauged basins is Stoll and Weiler's approach (2010), which estimates parameters of a distributed hydrological model (Hill-Vi) in ungauged basins by the explicit simulation of a stream network, and compares the simulated networks to mapped stream networks. They achieved some promising results but also encountered some limitations, such as the problems caused by simplification and modification of the model to achieve fast and robust runs. More recently, data-driven methods (e.g. ANNs, MLR, etc.) which are potentially applicable for streamflow prediction, have been investigated as alternative models to estimate streamflow in ungauged catchments.

In data-driven and conceptual/semi-distributed hydrological models, the model parameters have to be estimated through calibration against observed streamflow in the process of parameter adjustment (automatic or manual), until catchment and model behavior show a sufficient agreement. Since in ungauged basins an observed streamflow time series is not available, the

transposition of either gauged streamflow or model parameters from a similar and/or nearby gauged basin, called regionalization, is well recognized as a low-cost and popular solution to provide time series of streamflow at ungauged basins (Young 2006; Samuel et al. 2011).

The term “regionalization” has been used in the literature with almost identical concepts but some minor differences among writers. In Germany since 1993, the priority program “Regionalization in Hydrology” was funded by the German Science Foundation (DFG) was concluded in 1998. The conference on Regionalization in Hydrology held in 1997 aimed to explore new mathematical and computational tools to describe and analyze the behaviour of hydrological systems at all relevant scales, from the point to the global, for whole systems and subsystems. The priority program has identified spatial scaling as the main problem of transferring hydrological information between spatial objects and emphasized the need for the acquisition, management, analysis and cartographic presentation of large hydrological data sets for regionalization of hydrologic information, also application of the GIS technology is recognized as an essential prerequisite (Streit and Kleeberg 1996; Diekkruiger et al. 1999). Relevant research works occurred after 1998 have been affected by this program. For example, Hennrich et al. (1999) looked for regionalization techniques to transfer parameters from one scale to another while some other studies (e.g. Chiang et al. 2002; Mwale et al. 2011) used the term regionalization or “hydrologic regionalization” only for classifying the watersheds in terms of hydrologic characteristics, and later on used that classification for transferring hydrological information. Also, Mwale et al. (2011) regionalized runoff variability and accounted for runoff heterogeneity across Alberta, Canada using statistical methods and their further plan is to model the streamflow of clusters with several classes of hydrologic models.

According to Blöschl and Sivapalan (1995), regionalization refers to the process of transferring the information (i.e., “hydrological information”) from one catchment (location) to another, and it may be satisfactory if the catchments are similar (in some sense), but error-prone if they are not. Hydrological information can be either the model parameters or the general structure of models which estimate hydrological responses, e.g. streamflows. According to this definition, the ungauged catchments should be located in a region homogeneous with the gauged basins. The assumption behind the homogeneous region of runoff responses is that similar climate, geology, topography, vegetation, and soils in the homogeneous region would generally produce similar runoff responses, but not necessarily in geographically neighboring basins (Smakhtin 2001). In this paper, regionalization is considered as the process of transferring hydrological information from gauged to ungauged basins.

Regionalization can be used to transfer hydrological information between spatial objects, spatial dimensions (which classify the objects according to their number of independent directions in space), and scale (a general idea of looking at objects in a more or less generalized manner) (Streit and Kleeberg 1996). It can be done by using predictive approaches such as an extrapolation of hydrologic information from gauged to ungauged basins, observation by remote sensing and hydrologic model simulation, and integrated meteorological and hydrological modeling (Goswami et al. 2007).

Regionalization is widely regarded as a challenging task in hydrological science (Sivapalan et al. 2003; Oudin et al. 2008; Stoll and Weiler 2010; Samuel et al. 2011), firstly, because of the lack of runoff data which is normally used to calibrate parameters of hydrological models, limits the success of simulation (Sivapalan et al. 2003). Secondly the studies on regionalization methods usually produce different results, as they have been examined on

different sites and also the available catchment characteristics vary from one case to another (Oudin et al. 2008). As a result there is no universal method for regionalization.

Considering the rapid pace of regionalization-method development in the literature and the importance of continuous streamflow estimation in ungauged basins, we intend to present a comprehensive review of continuous streamflow regionalization approaches and their developments. Our overall motivations are to review regionalization approaches which have been developed and applied in estimating streamflow or runoff (streamflow per unit area of catchment) in ungauged basins; show the key steps in regionalization procedures, catchment attributes used in these studies, related uncertainty analysis and hydrologic model optimization; and finally present recent developments in the streamflow regionalization. He et al. (2011) have recently reviewed continuous streamflow regionalization methods, but they have only reviewed hydrologic model-dependent approaches. This paper provides a more comprehensive review including model-dependent and model-independent methods, and discusses the developments and emergent directions in continuous streamflow regionalization.

Main steps in the regionalization approach

In general, a regional model can be stated in a simplified form as follows Eq. (1) (Wagener and Wheater 2006):

$$\hat{\theta}_L = H_R(\theta_R|\Phi) + v_R \quad (1)$$

where $\hat{\theta}_L$ is the estimated hydrological variable of interest at the ungauged site (it can be an estimated model parameter, probability or cumulative distribution function parameter or hydrological response such as streamflow or flow events), $H_R(.)$ is a functional relation for $\hat{\theta}_L$ using a set of catchment attributes (physiographic and/or meteorological attributes - Φ), θ_R is a

set of regional hydrological variables of interests (e.g. model parameters); and v_R is an error term. Clearly, regionalization requires information of catchment attributes (physiographic and meteorological attributes) and a function (linear/nonlinear) for relating predictors to the predictand.

There are at least five main important steps in a regionalization strategy. The first step is to collect and manage catchment attributes, which includes meteorological attributes (such as mean annual rainfall, temperature, etc.) and physiographic information (such as the location of stream gauges or centroid of the drainage basins, area covered by grass, trees, etc., and soil types, permeability of soils, etc.). Unfortunately, the required information on catchment attributes is sometimes difficult to obtain. This raises the question on what minimum number and types of catchment attributes should be collected for a proper regionalization procedure, and also which regionalization methods are proper when all of the required catchment attributes are not available. In the next section different catchment attributes which are applied in streamflow regionalization studies are discussed.

The second step in a regionalization procedure is to determine and clarify hydrological variables of interests. In the case of continuous streamflow regionalization, the collected streamflow data of nearby and/or similar gauged basins will be used to generate hydrological model parameters or relationships with catchment attributes. The Third step is to develop a relationship between streamflow or runoff indices or hydrological model parameters and catchment attributes. This relationship is the regionalization method which will be discussed in detail in the next section. The fourth step is to evaluate model performance using pseudo-ungauged basins before the model is applied to the real ungauged basins. Once a relationship between catchment attributes and hydrological variables is established, it is necessary to validate the model before it can be

applied in ungauged basins. The leave-one-out cross-validation procedure is normally used to assess the validity of regionalization approach (e.g. Samuel et al. 2011; Parajka et al. 2005; Merz and Blöschl 2004). Other cross-validation techniques such as split-half and bootstrapped are not usually applicable to regionalization studies because they can only be applied to poorly gauged basins and for the periods in which the observed flow data is available. In leave-one out approach, each catchment is in turn considered as being ungauged for obtaining a flow simulation in that catchment, the actual discharges are afterwards used to evaluate the performance of the flow estimation procedures for the catchment. Statistical tests are then used to evaluate the performance of the flow estimation in the “pseudo” ungauged basins. Table 1 presents some error metrics commonly used. Among them are the Nash-Sutcliffe efficiency, Root Mean Square Error (RMSE), relative RMSE, mean bias (BIAS), relative BIAS, Volume Error (VE), and Correlation coefficient. The fifth and final step is to include uncertainty analysis. A major recent improvement in regionalization procedures is to include uncertainty analysis, which is necessary due to uncertainties in selecting catchment properties and regionalization procedures, and identifying gauged and regional model structures and their parameters (Wagener and Wheater 2006). A discussions and outline of the general uncertainty analysis procedure will be presented later on.

The main steps presented above provide an overview of the general procedure of the regionalization of any hydrological variable. Step 3 is where the regionalization model can be changed for different hydrological variables (e.g. select a different hydrologic model). Regionalization can be adjusted for specific purposes, such as linking to ecological implications (Stainton and Metcalfe 2007) or combining the approach with integrated data processing tools

such as Geographical Information Systems (GIS, e.g. Streit and Kleeberg, 1996; Cheng et al. 2006).

Catchment attributes

Catchment attributes are different across various studies, and it seems that an initial hypothetical judgment is required to identify which potential catchment attributes would have an impact on the runoff responses of interests. Merz and Blöchl (2004) and Parajka et al. (2005), for example, use the help of expert judgments to take into account the interaction between the runoff regime, climate, and physiographic attributes. Kokkonen et al. (2003) point out that catchment attributes used for regionalization purposes should characterize the factors that drive the hydrological response of a catchment and should also be derivable from existing and readily available data sources, such as topographical maps. Some researchers select catchment properties with the objective that the sample of catchments is representative of the statistical population of catchments in the region, both in geographical and in parameter space (Mwakalila 2003). Some might use step-wise regression analysis to identify landscape-climate descriptors that are good predictors of percentile flows (e.g. Mohamoud 2008) while some of the relationships identified by step-wise regression, even though statistically significant, might be an accident of the data (Sefton and Howarth 1998).

Table 2 presents the catchment attributes which have been used in reviewed papers. Types of catchment attributes collected to perform regionalization vary among studies. By simply comparing the types of data attributes used in a regionalization process it can be summarized that physiographic information (e.g. catchment area, elevation, slope of basins or channels) and meteorological attribute (e.g. mean annual or mean daily rainfall and temperature) are most often

incorporated in regionalization studies (either in regionalization or hydrologic model calibration). Other catchment attributes which are also occasionally incorporated in some studies are: (1) the percentage of area that is covered by water (e.g. lakes, swamp, wetland, and groundwater) or by land use (e.g. forest, grass, woodland, urban/non-urban area); (2) location of stream gauges or centroid of catchments; and (3) other meteorological attributes such as mean annual evaporation and snowfall. According to Mwakalila (2003), methods which are used to quantify the catchment attributes usually include topographical indices, geology and soil index, climate indices, and vegetation cover indices. Croke et al. (2004) collected catchment attributes on soil covers and physiographic characteristics, since their study focused on predicting hydrologic responses to land cover changes due to agricultural intensification in gauged and ungauged basins.

In general, the most widely used attributes by researchers in continuous streamflow regionalization are catchment area, elevation, slope of basins or channels, and mean annual or daily rainfall and temperature.

Developments in continuous streamflow regionalization from 1990-2011

Regionalization approaches can be classified into hydrologic model-dependent and hydrologic model-independent groups (Fig.1). The methods employed by the first group transfer rainfall-runoff model parameters between basins. Those model parameters are then used to generate continuous streamflow in the target basin. The second group does not estimate streamflow time series through rainfall-runoff models (in ungauged basins), and therefore instead of hydrologic model parameters, the equation structure and its parameters are transferred. These models usually develop and employ an equation representing input-output relationships, such as rainfall

and temperature as input and streamflow as output. Hydrologic models require more catchment attributes to be parameterized and more knowledge and expertise while the structure of data-driven models is simpler and can be defined with less data compared to the former. An interesting study has been the comparison of the two approaches. Goswami et al. (2007) have used both categories; conceptual hydrologic models and ANN models to simulate daily flow in 12 French catchments considered as ungauged. The result of their study indicated that ANN model had better performance. Although such comparative study requires more resources, it appears a good approach in identifying the optimum regionalization method for a case of interest. It is noteworthy that the good potential of data-driven methods (e.g. ANN) for streamflow regionalization can be particularly useful for cases where data is not available for applying hydrologic model.

There are about 70 published studies involving streamflow regionalization within the last two decades (which are cited in the paper). The summary of 43 representative studies is tabulated in Tables 3 and 4. Most of these regionalization studies estimate streamflow through hydrologic models and investigate the methods which transfer model parameters from gauged to ungauged basins (Table 3) while a lesser number use data-driven methods to infer streamflow directly in ungauged basins (Table 4). Table 3 presents the former category and contains references (authors' names and publication year), study area, climate classification of the study area, temporal scale and key objectives, name of the hydrologic model, regionalization methods which extrapolate the model parameters, and finally the major findings of the papers. Table 4 shows the latter category and indicates the hydrologic model-independent methods which regionalize streamflow from gauged to ungauged basins (see column "regionalization methods").

Tables 3 and 4 indicate that regionalization approaches have been widely used in many parts of the world such as North America, Asia, Australia, Africa, and Europe. The approaches have been applied to estimate streamflow from small basins (e.g. 0.04 km²) to large basins (e.g. 823000 km²) (see column “Study Area” in both tables). These approaches have also been applied to various landscapes, climatic regions, and different topographies. Parajka et al. (2005) and Merz and Blöschl (2004), for example applied the regionalization approaches in the catchments located in different topographies in Austria. Some examples of the streamflow regionalization approaches in both groups will be further described and discussed in the following sections.

Hydrologic model-dependent methods

The regionalization methods which use hydrologic models to estimate streamflow in ungauged basins are essentially methods which transfer the model parameters from gauged basins to ungauged ones. In recent studies, Peel and Blöschl (2011) and He et al. (2011) have reviewed the methods which are used to estimate model parameters of ungauged catchments. According to the classification of Peel and Blöschl (2011), these approaches are split into groups which estimate model parameters *a priori* (without calibration using relationships between model parameters and catchment characteristics), regionalize calibrated model parameters from gauged to ungauged basins (e.g. regression relationships, spatial proximity, and physical similarity), use multi-objective and regional calibration to assess predictive uncertainty, model (output) averaging, and hydrological signatures (indices) modeling. The described classification is a very general one, regardless of simulated hydrologic variables and includes the studies which aim to estimate uncertainty. The classification scheme employed by He et al. (2011) includes distance-based (using geographical and hydrological

similarities) and regression-based methods. This classification and the described examples, cover only the hydrologic-model dependent methods presented in the next paragraph.

In a more specific classification, we classify techniques which have been used to extrapolate hydrologic model parameters to estimate streamflow at ungauged basins into the following groups: (a) arithmetic mean method (e.g. Merz and Blöschl 2004; Oudin et al. 2008; Jin et al. 2009); (b) spatial proximity (spatial distance) approach (e.g. Merz and Blöschl 2004; Parajka et al. 2005; Oudin et al. 2008; Li et al. 2009); (c) physical similarity approach (Oudin et al. 2008; Samaniego et al. (2010) ; Samuel et al. 2011); (d) scaling relationships (e.g. Croke et al. 2004; Schreider et al. 2001); (e) regression-based methods (linear and nonlinear), such as ANNs, MLR (e.g. Sefton and Howarth 1998; Post and Jakeman 1999; Merz and Blöschl 2004 ; Parajka et al. 2005; Cheng et al. 2006; Young 2006; Heuvelmans et al. 2006; Götzinger and Bárdossy 2007; Oudin et al. 2008; Mohamoud 2008) (f) hydrological similarity approach (e.g. Masih et al. 2010) .

With techniques such as arithmetic mean and spatial proximity, in which catchment attributes are not directly involved, one assumes that all catchments within the particular radius are similar and differences in the parameter values arise only from random factors (as in studies performed by Merz and Blöschl 2004; Parajka et al. 2005). All techniques which transfer model parameters from one basin to another are based on the assumption that two basins are similar and will respond identically to the same input (Stoll and Weiler 2010). In the arithmetic mean approach, the rainfall-runoff model parameters of surrounding basins (local) or all basins (global) are averaged. Parajka et al. (2005), for example, used the arithmetic mean of parameters of a region within a radius of 50 kilometers from the catchment of interest for the local technique

and all basins for the global technique. Arithmetic mean might also be used as a computation technique after basin classification using other regionalization methods. Spatial proximity approaches transfer the model parameter sets based upon a spatial distance technique, i.e., an interpolation technique which is a function of the geographic location. The most popular interpolation technique in this context is kriging. Vandewiele and Elias (1995), applied a monthly water balance model to 75 Belgian basins which were considered as ungauged in turn and used two spatial proxy techniques (kriging and the use of parameter values of a few neighboring basins) to compute their parameter values. In their experience, kriging gave satisfactory results (root mean squared error less than 7% or relative error less than 20%) in 72% of the basins, whereas the second technique gave satisfactory results in only 44% of basins. Jin et al. (2009) used both of the described regionalization methods. They applied the HBV model using proxy-basin and global mean to regionalize model parameters in 13 sub-basins in southern China to simulate daily streamflow. With the global mean method, they constructed three different sets of regional mean parameters: an arithmetic mean regardless of the size and position of basins, area-weighted mean values of parameters, and a proxy-basin method. The parameter set of simple arithmetic mean values produced the best results compared to other applied methods.

The third method is a physical similarity approach. The concept of this approach is to transfer hydrological model parameters from gauged to ungauged basins according to the similarity of their physical attributes. An example of such an approach is presented by Oudin et al. (2008) and Samuel et al. (2011). In this method, catchments are first grouped according to their physical or non-hydrological similarities. Multivariate statistics analysis is normally used to group the catchments. It is recommended that one use a ranked proximity technique if catchment attributes

have different units and ranges. Then, rainfall-runoff model parameters of gauged basins are computed and model parameters located in the same group are arranged, for example by using an arithmetic mean, to obtain a regional rainfall-runoff model parameter set. That parameter set is then used to generate streamflow in the target basin having physical similarity. Some might use physical catchment attributes to find dissimilarity measures. For example, Samaniego et al. (2010) proposed a procedure to find a metric on the basis of dissimilarity measures that are estimated from pair wise empirical copula densities of runoff. They defined a metric in a transformed space of basin descriptors consisting of 22 physical characteristics of the basins, such as area, slope, elevation, permeability, imperviousness, and estimated streamflow in an ungauged basin by transferring parameters from gauged basins on the basis of the selected metric. Simulated daily discharge using their proposed methods has $NSE=0.76-0.86$, which is considered as a reasonable result. Reichl et al. (2009) selected an ensemble of hydrological models to optimize similarity measures among 27 geometric, climatic, soil and vegetative attributes of catchments to transfer SIMHYD model parameters to 184 Australian catchments and predict monthly streamflow. Their results indicate that flow prediction using an optimized model averaging method (based on physical similarities) is superior to regression and spatial proximity approaches.

The fourth method is scaling relationships based on area or other catchment attributes. For instance, Croke et al. (2004) computed model parameters in ungauged basins by scaling the area, deep drainage, and surface runoff of gauged basins according to area and estimates of surface runoff and deep drainage of ungauged basins obtained from other models e.g. an agricultural model. This method can be used for simulation of subcatchments' streamflow in ungauged subcatchments, assuming that the streamflow contribution from each subcatchment to the total

catchment yield is proportional to a ratio of the catchment area or other attributes (Schreider et al. 2002).

Another type of approach is the use of regression-based methods, which include nonlinear regression methods such as ANNs and linear regression methods. Within this group, a linear or nonlinear relationship is developed between model parameters and catchment attributes. Kokkonen et al. (2003) applied a regression scheme to the six-parameter IHACRES model and physical catchment descriptors (Table 2.) on thirteen Australian catchments to produce daily flow time series which provided to be more accurate than those produced by arithmetic mean method. Sefton and Howarth (1998) conducted a multiple regression analysis to find the relationship between IHACRES model parameters and the selected catchment attributes in 60 basins in the UK. They selected the important catchment attributes in defining hydrologic response by correlation matrix, step wise regression and principal component analysis. Some studies, such as those by Götzinger and Bárdossy (2007) and Cheng et al. (2006) used the multiple regression approach in their regionalization studies, albeit with some modifications due to the use of distributed hydrological model parameters in their model which results in large number of model parameters. Götzinger and Bárdossy (2007) used the transfer function approach to ensure consistent parameter estimation, whereas Cheng et al. (2006) used the multiple regression approach to govern the relationship between the areas of different land cover types occupying the drainage basins. Cutore et al. (2007) tested particular regionalization procedures to transfer the parameters of simple rainfall-runoff models, based on a “two-step” approach in which the first step is a simple regression-based between rainfall and streamflow and the second step is regression equations between model parameters and the geomorphological characteristics, such as average altitude, soil permeability, stream length, etc. The “one-step” approach is only

based on regional rainfall–streamflow model calibration for gauged basins. According to their experience models based on the “one-step” approach appeared to be robust and adequate for estimating the streamflow in ungauged basin.

The last method in this category considers hydrological similarities between basins to transfer the model parameters. A simple method based on hydrological similarity assumes that all catchments within the region are similar in their hydrological behavior. In this case, a mean of available or an entire set of calibrated parameter values is typically used to estimate the value in the ungauged catchment instead of deriving quantitative relationships between catchment descriptors and model parameters (Kokkonen et al. 2003; Mohamoud 2008). Flow duration curves (FDCs) similarity is also considered as a hydrological similarity which is used by Masih et al. (2010). They found that the HBV model parameters transferred from similar gauged basins based on FDCs’ similarity to a data-limited gauged catchment is a better basis for streamflow time series generation than similarity approaches based on drainage area and spatial proximity. This may be due to the fact that FDCs account for climatic factors.

Some studies resort to a combination of the aforementioned methods to transfer hydrologic model parameters. For example, after basin classification according to physical or hydrological similarities or spatial proximity criteria, regression-based or averaging techniques can be used to transfer the model parameters. More examples and details about reviewed studies in these categories are presented in Table 3.

The wide spectrum of regionalization studies uses hydrologic models and involves methods which transfer or estimate the hydrologic model parameters of ungauged basins using the gauged ones. Different approaches can produce different results in various regions. Using a cross-validation procedure and basing its results on model validation statistics, each study claims

that one method is more suitable than the others. But, in general, the best performing method is site-specific which seems driven by climate and physiographic conditions that play an important role in the differences of method performance from one region to another. Therefore a major challenge may lay ahead with climate change suggesting that an optimal regionalization method for a given basin may not be the appropriate one in the future. Further research is needed on that topic.

Hydrologic model-independent methods

As displayed in Fig.1, hydrologic model-independent (mainly data-driven methods) can be classified into three groups. The first group is regression-based analysis which includes linear regression (e.g. MLR, parametric simple linear model used by Goswami et al. 2007), and nonlinear regression (e.g. ANNs) between streamflow and catchment attributes. For instance, nonlinear regression methods are used in Besaw et al. (2010) and Parada & Liang (2010). The former trained the recurrent ANNs on climate-flow data from one basin and used them to forecast streamflow in a nearby basin with different (more representative) climate inputs, and found that ANNs that always converge and avoid stochastic training (e.g. General Regression Neural Network (GRNN) and Counter Propagation Neural Network (CPNN)) are straightforward to execute and widely applicable to small ungauged basins. The latter presented a Kernel-based methodology, which is applicable to those ungauged basins whose hydrological and system properties/behavior are non-linear and non-Gaussian and inferred streamflows for two basins in the US treated as ungauged. The next category is time series models, e.g. AutoRegressive (AR) models. Chiang et al. (2002a) used Multiple Regression Analysis (MRA) to construct relationships between the parameters of streamflow time series (dependent variables) and watershed characteristics (independent variables). The predicted streamflow

parameters from the MRA equations were then used to synthesize hydrographs under the time series model. The regression equations might be developed between streamflow percentile and catchment attributes. For example, Mohamoud (2008) developed a linear regression between the parameters of a FDC model and landscape-climate descriptors identified through a step-wise regression method for gauged basins, transferred these equations to ungauged basins, and converted the simulated FDCs to a streamflow time series. Isik and Singh (2008) also used FDC models to compute streamflow discharge at ungauged sites in 26 river basins in Turkey after defining homogeneous regions with a hierarchical clustering algorithm. They found that FDCs constructed for each homogeneous region estimated monthly discharge data with correlation coefficients of 95-100% for most of the gauging stations. The last category of these approaches is scaling methods. Buttle and Eimers' (2009) study on the Precambrian Shield in south-central Ontario indicates that annual maximum and mean daily streamflow can be well-predicted using simple scaling relationships. These scaling relationships between streamflow metrics and basin size are potentially capable of the extrapolation of data from gauged basins to ungauged sites (Yue and Wang 2004). A similar study on the Mackenzie River Basin in northern Canada estimated streamflow for ungauged basins using streamflow-area relationships of gauged basins (Woo and Thorne 2003). In addition to streamflow time series, the statistics of these time series are also potentially applicable in scaling analysis. Lima and Lall (2010) remarked that statistics of the streamflow time series also scale with physical properties of the drainage basin such as catchment area. More examples of the second category of regionalization methods are presented in Table 4.

The accuracy of predictions can be impacted by time scale, i.e., the time step of prediction based on catchment area. In Besaw et al. (2010) study which examines the case of streamflow

prediction at two small sub basins of about 2,700 km², predictions using hourly data were slightly more accurate than those using daily data (Root Mean Square Error of 5.5 versus 5.2) . However such minor difference in model performance may not allow drawing conclusion on the time scale effect on model performance. More studies are needed to clarify that issue.

In conclusion, among the hydrologic model-independent methods for streamflow regionalization described above, regression equations (including linear and nonlinear) developed between the desired hydrologic responses, (e.g. streamflow) and catchment attributes are the most commonly used ones. One of the advantages of hydrologic model-independent methods is the lower data requirement of these approaches and the simplicity of their structure which does not require special knowledge and expertise of hydrological modeling. However, the identification of an appropriate model structure (e.g. ANN architecture) requires some expertise. Data-driven methods do not simulate the actual rainfall-runoff process and therefore they are not impacted by uncertainty due to the physical process being modeled; however, they are still affected by other sources of uncertainty, e.g. the estimation method and its parameterization.

Hydrologic models in streamflow estimation in ungauged basins

Compared to conceptual and semi-distributed models (e.g. SAC-SMA), fewer studies have used fully distributed physically-based models (e.g. MIKE SHE) to estimate streamflow time series in ungauged basins. Using fully distributed physically-based models for streamflow estimation in ungauged basins is not usually involved with regionalization process due to *a priori* parameter calibration. However, Gotzinger and Bardossy (2007) used a grid-based modification of the HBV model concept for simulation of catchment runoff in ungauged basins using regionalization approaches. While majority of studies calibrate them *a priori* . For example, Ibrahim and Cordery (1995) developed a model which represents hydrological processes in a physically

realistic manner to estimate runoff volumes from generally available geophysical data of ungauged basins. The inputs to the model were monthly rainfall, monthly climate data, and reference soil characteristics. The model was able to provide good estimates of volumes of flow on ungauged catchments in New South Wales. Nyabeze (2005) estimated runoff using a distributed parameter value estimation approach for various poorly gauged catchment sizes in Zimbabwe, and simulated runoff of different segments independently, in which each segment represents an area of different rainfall, soil condition, vegetation cover, and land use, which all influence runoff. Ao et al. (2006) proposed a mixed model structure BTOPMC (Block wise use of TOPMODEL with Muskingum-Cunge flow routing method), which is a physically-based model that has the advantages of both lumped and distributed models to avoid large data requirements, uncertainty and high cost, which often limit the distributed model's applicability. Their experience finally resulted in a reasonable agreement between the simulated and recorded runoff. The application of more detailed physically-based hydrologic models along with regionalization methods appear to yield some improvement but at a quite high cost. This approach is data intensive, and may not be appropriate in data poor regions.

In hydrologic model-dependent regionalization methods (Table 3), a rainfall-runoff model is first selected (see column "Hydrologic Model"). The rainfall-runoff models selected in regionalization studies are usually conceptual and semi-distributed models such as HBV (e.g. Jin et al. (2009)), IHACRES (e.g. Young 2006), HYMOD (e.g. Castiglioni et al. 2010), RRMT (e.g. Wagener and Wheater 2006), SIMHYD (e.g. Peel et al. 2000), PDM (McIntyre et al. 2005), AWBM (e.g. Boughton and Chiew 2006), Xinanjiang (e.g. Li et al. 2009), GR4J and TOPMO (e.g. Oudin et al. 2008), SMHI (Hundecha et al. 2008), SWAT (e.g. Heuvelmans et al. 2006)

and TOPMODEL (e.g. Bastola et al. 2008) and rarely physically-based models such as Mike 11 NAM (Makungo et al. 2010) . The full names of these models can be found in Table 3.

In selecting an appropriate model, there is a consideration of data-availability and that the model should be simple (parsimonious with less model parameters), able to capture more hydrological variability and behaviors required in the study catchments (such as snowfall, lakes, etc.), and has been effectively applied in similar regions. In the regionalization approach, a parsimonious hydrologic model structure is usually selected and calibrated to observable watershed responses for a large number of gauged watersheds (Yadav et al 2007). Next step is the selection of regionalization model to govern the relationships between catchment attributes and model parameters in gauged basins and transfer them to ungauged basins.

The most widely used hydrologic models in streamflow regionalization studies are HBV and IHACRES. This may be due to the fact that those models represent well the key components of rainfall-runoff process without unnecessary details. To identify the best model for streamflow estimation in ungauged basins in a certain region, both physically-based and conceptual hydrologic models are suggested to be examined. Makungo et al. (2010) applied both Mike 11 NAM (a physically-based model) and AWBM (a conceptual model) for estimation of daily discharge in ungauged basins of a catchment in South Africa using regionalization method of spatial proximity and found that AWBM slightly outperforms the other model. But, in practice, usually both categories of models available for streamflow estimation in ungauged basins are not examined for various reasons. In future studies, the benefit of using methods from each category should be further emphasized.

Model parameter calibration (optimization process)

In conceptual and semi-distributed rainfall-runoff models, most parameters are not measurable and have to be estimated by calibration using observed runoff data. Optimized model parameters are required in model parameter regionalization, since the most common approach in this context is to look for relationships between optimized parameter values and catchment characteristics in gauged basins. Parameter sets can then be compiled for ungauged catchments from measurable variables (Seibert 1999). The range of parameter values is usually selected based on authors' knowledge of the study area and the experience of other studies (Seibert 1999) as well as initial model runs for the study area. Recent advances in optimizing rainfall-runoff model calibration have focused mainly on incorporating multiple objective measures of model performance and improving optimization algorithms (Li et al. 2010). Some examples of model parameter optimization efforts in the reviewed studies are presented in this section.

Oudin et al. (2008) used local gradient search procedure as an optimization algorithm for GR4J and TOPMODEL, with four and six parameters respectively on gauged catchments. Li et al. (2010) combined multi-objective optimization with averaging across multiple calibration sites and estimated model parameters by multi-objective optimization at each calibration site, and then finalized it by weighted averaging of the parameter values across sites. The weight for each site was calculated from the prediction error at that site. Mwakalila (2003) calibrated the Data-Based Mechanic (DBM) model using the Generalized Likelihood Uncertainty Estimation framework whereby, using the Monte Carlo simulation technique, the model was run 5,000 times with different randomly chosen parameter sets. The feasible range of parameters to be sampled was based on the initial estimated parameter values. The optimum values of parameter sets for each catchment were selected through an evaluation of the likelihood measure. Kim et al. (2008) used the large scale trust-region method to solve nonlinear large scale optimization problems. Li et al.

(2009) used the Particle Swarm Optimization (PSO) to optimize the model parameters. PSO is a relatively new addition to the evolutionary computation methodology and is expected to provide the global or near-global optimum. Samuel et al. (2011) used Brent's parabolic interpolation method to optimize the MAC-HBV model parameters.

In most of the mentioned optimization algorithms, multiple objective measures of model performance are used to improve the optimization algorithm and pick the least uncertain set of model parameters.

Developments in continuous streamflow regionalization with uncertainty estimation

In 2003, PUB focused on the estimation and subsequent reduction of predictive uncertainty as its central theme (Sivapalan et al. 2003). In regionalization studies, the inclusion of uncertainty analysis is necessary, firstly because model uncertainty is inherent and unavoidable (Yadav et al. 2007) and secondly because there are uncertainties in selecting catchment properties and regionalization procedures, and identifying gauged and regional model structures and their parameters (Wagener and Wheeler 2006). In addition, calibration procedure to obtain regional model parameters increases the uncertainties (Wagener and Wheeler 2006; Yadav et al. 2007; Zhang et al. 2008; Samaniego et al. 2010). Clearly, uncertainty analysis needs to be incorporated within the regionalization procedure to provide better predictions of streamflow in ungauged basins. However, this has been a challenging task over the last decade because of the various sources of uncertainties involved.

Some examples of studies applying uncertainty analysis in regionalization include: Wagener and Wheeler (2006); Yadav et al. (2007); Zhang et al. (2008); Samaniego et al. (2010); and Masih et al. (2010). Briefly, the methodology of uncertainty analysis is performed by calculating

prediction limits, determining upper and lower bounds for each parameter, simulating Monte Carlo or any other multi objective function tools to obtain behavioral or non behavioral parameter sets, and then using a behavioral simulation to create ensemble predictions in the ungauged basins. Wagener and Wheeler (2006) detected the uncertainties that resulted from difficulties in identifying model parameters, finding an appropriate calibration strategy and model structure errors by investigating the role of each model parameter and model structure behaviors' effects on streamflows in ungauged basins. Yadav et al. (2007) used a Monte Carlo simulation, whereas Zhang et al. (2008) applied the ϵ -NSGAI algorithm. Zhang et al. (2008) claimed that the use of the ϵ -NSGAI is more efficient and robust in finding behavioral parameter sets, and it can reduce uncertainty bounds of streamflows in ungauged basins. McIntyre (2005) regionalized conceptual rainfall-runoff models on the basis of ensemble modeling and model averaging. They remarked that the ensemble of candidate models provides an indication of uncertainty in ungauged catchment predictions, although this is not a robust estimate of possible flow ranges, and frequently fails to encompass flow peaks. Samaniego et al. (2010) accounted for uncertainty of parameterization by running the hydrologic model in an ungauged basin with sets of global parameters obtained from the k nearest neighboring donor basins using various metrics. With 22 basin descriptors, a Monte Carlo experiment was carried out to assess the explanatory power of possible combinations of these predictors. Masih et al. (2010) assessed the impact of parameter uncertainty on the regionalization results using the best parameter set of a study catchment in the regionalization procedure among 50 different parameter sets of a catchment that yielded in the highest Nash Sutcliffe Efficiency (NSE) values during the automatic Genetic Algorithm (GA)-based optimization procedure while Heuvelmans et al. (2006) assessed the uncertainty of the regionalization procedures with a non-parametric

bootstrap method. Samuel et al. (2011) calculated a measure of the goodness of fit between the observed streamflow and model predictions (NSE) for each parameter combination in each basin. If the goodness-of-fit value for any parameter combination was inside the accepted range, the parameter set was accepted to provide flow-prediction confidence limits.

An effective approach to reduce the impacts of parameter uncertainties on model simulations might be an appropriate reduction of the number of model parameters as is found by Huang and Liang (2006). They introduced a subsurface flow parameterization based on the concepts of kinematic wave and hydrologic similarity, with one parameter for calibration into VIC-3L model to reduce the impacts of model parameter uncertainties on model simulations by reducing the number of model parameters that need to be estimated through a calibration process. More recently, Samuel et al. (2011) resorted to a Monte Carlo simulation technique to capture flow variability in ungauged basins in Ontario. Significant developments in streamflow regionalization with uncertainty estimates have been achieved in the last decade. However, there is still no general method that could be applied to streamflow regionalization. This is in part due to the variety of the regionalization methods, and the different level of details of hydrologic models and data involved. This remains an interesting research avenue specially in a context of changing climate and land use/cover.

Discussion and Conclusion

This paper provides a comprehensive review of regionalization methods for estimating continuous streamflow or runoff in ungauged basins. Continuous streamflow regionalization can be carried out through a hydrologic model in which its model parameters are used as instruments to transfer hydrological information from gauged to ungauged basins. An alternative method is the use of hydrologic model-independent approaches which are usually based on data-driven

methods. The use of rainfall-runoff models causes some sources of uncertainties due to errors in computing local and regional model parameters and the relationship between local parameters and catchment attributes, and due to the uniqueness of the catchments (Wagener and Wheeler 2006) and the structure of the rainfall-runoff model. Conversely, hydrologic model-independent methods such as data-driven models avoid the impact of hydrologic model structure and parameter uncertainty, but still contain uncertainties due to the estimation method which is derived from the gauged basin and uncertainty due to the data-driven model's calibration. Data-driven methods usually require less data and expertise which might be useful to deal with constraints of data-availability. There is no clear indication from the literature on which regionalization approach should be preferred for a given case. Comparative study using methods from each category (model-independent and model-dependent) appears the best approach to identify appropriate regionalization method for a given site or region.

Since the regionalization process is inherently involved with using catchment attributes, it is not possible to establish a universal approach as the best method for all of the catchments. Therefore a specific study needs to be done on any region of interest to identify the best approach among hydrologic model dependent or independent methods. For example most reviewed studies on hydrologic model-dependent methods in arid to warm temperate climate (e.g. Australia) indicate that physical similarity and spatial proximity appears to be the best approach, while in warm temperate (most European countries) regression-based methods have been preferred. Similarly, in cold and snowy climate (e.g. Canada) spatial proximity and physical similarity approaches seem to outperform other hydrologic model-dependent methods. It has been found that the HBV and IHACRES are the most frequently used hydrologic models. Among the hydrologic model-independent methods, linear and nonlinear regression methods

have performed well in warm temperate regions (e.g. European countries) while in cold and snowy climate (e.g. Canada) and warm humid climate (e.g. Brazil) scaling relationships have shown good performance. This highlights the site-specific nature of the regionalization methods and the need for comparative study before selecting a regionalization method for a given site or region. Existing studies summarized in this review can provide some guidance and reduce the number of methods to be investigated.

Wagener and Wheater (2006), and later followed by other researchers such as Yadav et al. (2007), Zhang et al. (2008), Besaw et al. (2010), and Samaniego et al. (2010) suggested and then applied uncertainty analysis in their regionalization studies. The use and development of the regionalization approach, including uncertainty estimate, is in line with the central theme of the PUB initiative promoted by the International Association of Hydrological Sciences (IAHS, see Sivapalan et al. 2003). Although significant developments have been achieved on that topic, in general, there is still no standard method for estimating uncertainty in streamflow estimation at ungauged basins using regionalization techniques. This issue remains a challenging research topic in a context of climate change along with changes in land use and cover. Thus it is raising the emerging issue of nonstationarity that is generally overlooked by most regionalization methods that assume stationarity. Future research should focus on those key issues to advance the regionalization techniques for streamflow estimation in ungauged basins.

There has been some preliminary work in studying the nonstationary effects on streamflow prediction in ungauged basins. For instance, Lima and Lall (2010) developed and applied hierarchical Bayesian models, to assess regional and at-site trends in time in a spatial scaling framework and to reduce parameter and model uncertainties. They used a log-log scaling law of annual maximum series and catchment area to estimate varying scaling parameters.

This review provides a comprehensive overview of various methodologies used in continuous streamflow regionalization over the last two decades. Clearly, after 1990 hydrologic variables regionalization approaches have remarkably progressed. Recent improvements in this area include for example the use of data-driven methods in both hydrologic model's parameters extrapolation and hydrologic response inference such as the ANN approach proposed by Heuvelmans et al (2006), Goswami et al. (2007), Besaw et al. (2010); and the statistical inference methods for watersheds with non-linear behavior (Parada and Liang 2010); the transfer function approach (Göttinger and Bárdossy 2007); the use of uncertainty analysis with an optimum multi-objective function (Zhang et al. 2008); nonstationarity analysis (Lima and Lall 2010), and the incorporation of multi-objective measures of model performance and improved optimization algorithms for watershed models calibration (Li et al. 2010).

The rapid development of streamflow regionalization, in terms of its methodologies after 1990 and applications as shown in this paper brings greater confidence that accurate streamflow estimation at ungauged basins can be achieved. However, there are still many particular aspects that require more research to improve regionalization analysis and results. For example the problem of data-availability raises questions about which approaches of streamflow estimation in ungauged basins should be taken. When a rainfall-runoff model is to be selected, which model is most efficient and applicable and also parameter calibration or optimization approach needs to be addressed. Uncertainty analysis is always required and strongly recommended in regionalization efforts; however a well established method for uncertainty estimation in the context of streamflow regionalization is still to be developed. The emerging issue of nonstationarity in hydrological time series (e.g. streamflow time series) due to climate change and changes in land use and cover, raises question as to the validity of most of the regionalization methods used with

the assumption of stationary. Significant work still lies ahead to properly address both the issue of “nonstationarity” and “uncertainty estimate” in the context of streamflow regionalization.

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Table 1. Validation tests (error metrics) which are normally used for regionalization studies

Validation test	Equation	Note of variables in the equation
Nash-Sutcliffe efficiency	$NSE = 1 - \left(\frac{\sum_{i=1}^N (y_i - y'_i)^2}{\sum_{i=1}^N (y_i - y_{mean})^2} \right)$	
Volume Error	$VE = \frac{\sum_{i=1}^N y'_i - \sum_{i=1}^N y_i}{\sum_{i=1}^N y_i}$	y_i = observed streamflow at time-step i y'_i =modeled streamflow at time-step i y_{mean} = the mean of the observed values N = number of time-steps (data points)
Mean Bias	$BIAS = \sum_{i=1}^N (y'_i - y_i)$	
Mean Relative Bias	$BIASr = \sum_{i=1}^N \frac{(y'_i - y_i)}{y_i}$	

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Table 2. Catchment attributes used in streamflow regionalization studies

References	Catchment attributes	References	Catchment attributes
Vandewiele and Elias 1995	Location of basins	Merz and Blöschl 2004	Area, elevation, slope, porous aquifers, land covers, geologic units, soil types, river network density, lake index, mean annual precipitation, maximum annual daily precipitation
Sefton and Howarth 1998	Morphometric (elevation, area, channel slope, etc); Soils (% groundwater, %shallow groundwater, % peaty soil, etc); Land use (%grass heath, mixed woodland), rainfall and temperature	Lee et al. 2006	Daily precipitation, streamflow and potential evaporation, 17 physical catchment characteristics e.g. catchment drainage area, altitude, slope
Post and Jakeman 1999	Area, elongation, slope of lid, gradient and drainage density	McIntyre et al. 2005	Daily precipitation, streamflow and potential evaporation, drainage area , standard- period average annual rainfall , mean catchment altitude (m above sea level),index of fractional urban extent
Siebert 1999	Area, % forest, % field of meadow, and % lake	Parajka et al. 2005	Similar as Merz and Blöschl (2004)
Peel et al. 2000	Climate, terrain (90th percentile minus 10th percentile elevation in catchment), soil depth, and plant water holding capacity	Boughton and Chiew 2006	Rainfall, runoff , median elevation ,elevation range , leaf area index; percent of woody vegetation on catchment; plant water holding capacity, transmissivity
Chiang et al. 2002	Watershed area, forest area, percentage of contributing drainage area, area of storage, elevation, above mean sea level, stream length per unit area, main channel slope, the mean annual precipitation	Cheng et al. 2006	Land covers: vegetated area (agriculture & natural forest), bare field, vegetation and urban area
Kokkonen et al. 2003	Mean overland flow distance to a stream , mean flow distance in a stream , mean solar radiation index , mean topographic wetness index , catchment area ,elevation at the weir , mean catchment slope	Heuvelmans et al. 2006	Area, slope, height (masl), dominant land use, dominant soil texture
Mwakalila 2003	Catchment area and slope , drainage density and land use types, indexing topography, geology, and land use, mean annual rainfall (RAIN) and potential evapotranspiration	Wagener and Wheeler 2006	Baseflow index derived using the HOST classification, soil moisture deficit, catchment size and drainage path configuration, catchment steepness, dominant aspect of catchment slopes, average annual rainfall, median annual maximum 2-day rainfall
Croke et al. 2004	Forest cover area, catchment area, soil and topography classes	Young 2006	Catchment area, Mean catchment altitude, The mean aspect of direction of all slopes, The longest drainage path, etc.
Cutore et al. 2007	Area, average altitude (masl) , permeable area (%) , stream length , record size	Post 2009	Daily rainfall ,mean average wet season rainfall, total length of streams, percent cropping and percent forest in the catchment , minor till (>1 m deep) ,thin till, rock ridges peat pond bedrock outcrops
Goswami et al. 2007	Area, length of longest stream, altitude at outlet and the highest point, mean latitude and hydrological variables	Reichl et al. 2009	Geomorphic characteristics e.g. min / max/ mean elevation, Climate e.g. mean winter & summer precipitation, Soils e.g. mean soil depth mean plant available water holding capacity etc.
Götzinger and Bárdossy 2007	Flow time, land use, soil properties, area and geology	Seibert and Beven 2009	The monthly long-term mean potential evaporation, the areal, corrected precipitation, Temperature
Yadav et al. 2007	Dynamic response characteristics in seven categories , magnitude of high flows, low , average flows, duration of flows, frequency, rate of change in flows, and timing of flow events & and 13 physical watershed characteristics e.g. climate, watershed topography and subsurface geology and soils	Zhang et al. 2008	Maximum & minimum temperature, incoming solar radiation, actual vapor pressure and precipitation, potential ET, area, aridity index, mean elevation, mean slope in degree, stream length, mean Solum thick, water holding capacity, Mean woody vegetation fraction

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Table 2. Continued

References	Catchment attributes	References	Catchment attributes
Bastola et al. 2008	Rainfall, Potential Evapotranspiration (PET) and runoff data	Besaw et al. 2010	precipitation and temperature , streamflow time series
Hundecha et al. 2008	Percentages of the different land use and soil classes, size and mean slope of the subwatershed, a shape factor defined as the ratio between area of a subwatershed and the square of the distance from the outlet of the watershed to the farthest point in the watershed.	Bocchiola et al. 2010	Glaciers meteorological data including temperature, precipitation and snow and ice ablation.
Kim and Kaluarachchi 2008	Drainage area , soil depth , percent of cultivated area , percent of forest area, mean Channel density , slope of main channel , ratio of main stream length to basin length , ratio of basin width to basin length , saturated hydraulic conductivity of the upper zone , precipitation of the wet season, dry precipitation of the non-wet season, mean annual potential evapotranspiration , total channel length	Castiglioni et al. 2010	Drainage area; percentage of permeable area; maximum, mean and minimum elevations, concentration time; mean annual precipitation; mean annual temperature
Mohamoud 2008	Land use and land cover%, Geomorphology e.g. minimum elevation, basin relief, average slope, etc. Soil e.g. total soil depth, geology: e.g. dominant lithology	Lima and Lall 2010	Drainage area, flood and streamflow time series
Oudin et al. 2008	Catchment area, mean slope, median altitude, river network density, fraction of forest cover, aridity index (E/P)	Makungo et al. 2010	Rainfall, evaporation, uncontrolled spills, downstream flow releases, dam water levels , domestic abstractions, area under irrigation, crop factors, types of crops grown and the irrigation schedule.
Buttle and Eimers's 2009	Physiographic characteristics e.g. area (ha), mean slope (%), drainage density (m) , fraction of basin area	Parada & Liang 2010	Historical streamflow records for gauged basins in the proximity of the ungauged watershed and streamflow prediction for ungauged basins, mean annual precipitation, area, location of outlet, elevation
Li et al. 2009	Hydro climatic characteristics e.g. mean annual precipitation , mean annual potential evapotranspiration , mean annual runoff , Runoff coefficient ,ratio of potential evapotranspiration to precipitation, catchment slope, plant available water capacity of soil, fraction of total woody vegetation	Samaniego et al. 2010	Area, mean & median slope , drainage density, shape factor , percentage of north and south facing slopes (%) , mean elevation (m) , saturated areas (%), mean available soil water capacity (mm) , fraction karstic formations (%) ,mean percentage of forest covered areas (%) , mean percentage of impervious areas mean percentage of permeable areas (%), mean annual precipitation (mm) , mean & maximum temperature
Jin et al. 2009	Area and position of basins	Samuel et al. 2011	The location of the centroid of the catchments (i.e., latitude and longitude), the morphology of the catchments (i.e., mean elevation, mean catchment slope and area), the percentage of area covered by water , the land use , water drainage, rooting depth and the surface geology , precipitation and temperature
		Bulygina et al. 2011	Hourly records of precipitation, incoming solar and net radiation, wet and dry bulb temperature, and wind speed and direction, area, mean channel slope, forest%, soil (HOST)%

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Table 3. Studies on hydrologic model-dependent streamflow regionalization approaches (1990-2011)

References	Study area	Climate*	Temporal scale and key objectives	Hydrological model	Regionalization methods	Class**	Major findings
Vandewiele and Elias 1995	75 Belgian basins (19-1597 Km ²)	Warm temperate	To obtain the monthly water balance of ungauged catchments	A monthly water balance model obtained by geographical regionalization	Spatial Proximity (kriging and the use of few neighbouring basins)	2	Kriging is acceptable (RMSE<7%) in 72% of the basins, whereas the second is acceptable in 44%.
Sefton and Howarth 1998	60 catchments in England and Wales (8.7-893.6 Km ²)	Warm temperate	To estimate daily flows in ungauged basins	IHACRES with GIS databases	Multiple regression	5	The relationships are limited in terms of statistical accuracy but robust enough to reproduce daily flows, NSE = 0.56 - 0.72
Post and Jakeman 1999	17 small catchments (4 to 65 (ha)) Victoria, Australia	Warm temperate	To estimate daily continuous flow in ungauged basins	IHACRES	Multiple regression (linear & nonlinear)	5	NSE around 0.72, errors in the representation of hydrologic response by the model and the limited availability of appropriate landscape attributes caused some poor results
Siebert 1999	11 catchments in flat regions (7 to 950 km ²) Central Sweden	Snowy	To develop a model to obtain optimized hydrological parameters and daily hydrograph simulation	HBV (plus Monte Carlo simulation to obtain optimized parameters)	Regression equations (linear, exponential, power and log)	5	Observed and simulated hydrograph : R _{eff} values: 0.70-0.88- fuzzy measure=0.85, About half of the parameters were significantly correlated to catchment characteristics
Peel et al. 2000	331 catchments (50 to 2000 km ²) in Eastern & southwestern Australia	Warm temperate	To determine the relationships between hydrological model parameters and catchment attributes and estimate daily streamflow in ungauged basins	SIMHYD	Linear regression between model parameters and catchment attributes	5	The correlations between the model parameters and the climate index are more statistically significant compared to other attributes (NSE : 0.12-0.24))
Schreider et al. 2002	Gauged and ungauged subcatchments of Thailand (68- 2157 km ²)	Equatorial	Prediction of monthly discharge in ungauged basins	IHACRES	Disaggregation procedure using measured streamflow data from a larger gauged catchment in which the ungauged sub catchments may be nested	4	Relative error for the monthly time step: 13–17%
Kokkonen et al. 2003	13 catchments in North Carolina, USA (1626 ha)	Warm temperate	Investigation of regionalization approaches for daily streamflow prediction	IHACRES	Arithmetic mean, regression, similar hydrological behavior	1,5,6	Arithmetic mean more often gives poor results in comparison with other methods; Elevation was the most powerful explanatory variable in the regressions
Croke et al. 2004	3 subcatchments in northern mountainous regions of Thailand (68- 2157 Km ²)	Equatorial	Prediction of hydrologic response to land use changes by simulating daily flow in ungauged catchments	IHACRES combined with CATCHCROP (conceptual crop model)	Scaling of parameters according to the ratio of the area of the gauged to ungauged catchments	4	The procedure was able to predict the relative pattern of annual and seasonal flows- Bias= 0.4-0.18 %, NSE<=0.85
Merz and Blöschl 2004	308 catchments in Austria (3 to 5000 km ²)	Arid and warm temperate	Estimation of continuous daily flow in ungauged basins	HBV	Multiple regression	5	All model parameters are associated with some uncertainty-NSE=0.63-0.67

*Climate classification according to Köppen-Geiger climate classification, source: <http://koeppen-geiger.vu-wien.ac.at/present.htm>

** Class is according to the predefined classes of approaches in Fig.1.

NSE: Nash and Sutcliffe efficiency

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Table 3. Continued

References	Study area	Climate	Temporal scale and key objectives	Hydrological model	Regionalization methods	Class	Major findings
Lee et al. 2006	131 catchments in UK (1-1700Km ²)	Warm temperate	Prediction of daily runoff in ungauged catchments	Aversion of PDM of Moore & a routing module	A step-wise multiple regression	5	Loss in NSE less than 0.04 in 90% of catchments (regressed model parameter values and locally optimized set)
McIntyre et al. 2005	127 catchments in UK (1-1700 km ²)	Warm temperate	Predictions of daily runoff in ungauged catchments	Probability distribution model (PDM) of Moore	Regression; ensemble modeling and similarity weighted averaging (SWA) method	5,3	Ensemble modeling & SWA method provides the best results and performs significantly
Parajka et al. 2005	320 catchments in Austria (10-9770 km ²)	Arid and warm temperate	Regionalization of catchment model parameters to produce daily runoff	HBV (lumped model but allowing different model states in different elevation)	Arithmetic mean (global and local mean), spatial proximity, regression and physical similarity	1,2,3,5	Spatial proximity (Kriging) and physical similarity perform best; NSE = 0.62 and 0.66
Boughton and Chiew 2006	213 catchments in Australia (50-2000 km ²)	Arid and warm temperate	Estimation of average annual runoff	Australian Water Balance Model (AWBM)	Multiple linear regressions	5	Two-thirds of the estimates of average annual runoff were within 25% of the actual value. NSE=0.3 to 0.97
Cheng et al. 2006	18 drainage basin, Great Toronto Area (6000Km ²)	Snowy	Predicting daily runoff for ungauged basins using gauged basins and precipitation data	SCS curve number (CN) and model and IHACRES	Multiple regression	5	The SCS curve model is more parsimonious than the IHACRES
Heuvelmans et al. 2006	25 catchments in Belgium (2.24-209 Km ²)	Warm temperate	Daily stream flow simulation in ungauged basins	SWAT (Soil and Water Assessment Tool)	ANN-based and linear regression	5	ANNs delivered more accurate parameter estimates than linear regression
Wagener and Wheeler 2006	10 catchments, Southeast of England (28.5-1261 Km ²)	Warm temperate	Investigation of uncertainty of rainfall-runoff model parameters and structure in streamflow regionalization	Rainfall-Runoff Modeling Toolbox (RRMT)	Linear and multivariate regression analysis	5	The uncertainty in the locally estimated model parameters is a function of their importance in representing the response of a given catchment
Young 2006	260 catchments within the UK (median area=110.25K m ²)	Warm temperate	Estimation of daily streamflow in ungauged catchment	IHACRES	A regression-based approach ; nearest neighbour based approach	2,5	The regression-based approach yields the best predictive results
Cutore et al. 2007	9 sub-basins in Italy (47-1832Km)	Warm temperate	Monthly runoff prediction in ungauged basins	Simple rainfall-runoff models	Regression-based approaches :one-step and two step, ANN	5	The “one-step” approach appear to be robust and adequate for estimating the streamflow in ungauged basins
Götzinger and Bárdossy 2007	The Nectar catchment in Germany (14,000 km ²)	Warm temperate	Estimation of continuous flow in ungauged basins	Distributed HBV model (the use of square grid cells as primary hydrological units)	Transfer function (linear of log relationships between model and transfer function parameters), Modified Lipschitz, monotony condition and combination of modified Lipschitz and Monotony condition	5	Best NSE for Lipschitz condition =0.5 and for 3 other methods 0.47

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Table 3. Continued

References	Study area	Climate	Temporal scale and key objectives	Hydrological model	Regionalization methods	Class	Major findings
Bocchiola et al. 2010	Pantano basin in the Italian Alps (11 km ²)	Warm temperate	Evaluation of the daily flow discharges	A developed model contains snow and ice melt calculation and semi distributed flow routing	Regression	5	Nash–Sutcliffe efficiency=0.67-0.75, results show a somewhat good capability of the model to evaluate the average flow discharge within the stream
Castiglioni et al. 2010	52 catchments located in northern central Italy (14-3082 Km ²)	Warm temperate	Parameter estimation for rainfall-runoff models and daily streamflow estimation	HYMOD	Regional calibration using a maximum likelihood function based on regression relationship	5	Approximate likelihood cannot take into account the low frequency behaviors (long term autocorrelation), Nash–Sutcliffe efficiency=0.22-0.67, the results are inferior to classical calibration but it is encouraging
Masih et al. 2010	Karkheh river basin, Western part of Iran (50,764 km ²)	Arid	Simulation of daily streamflow for poorly gauged catchments based on hydrological similarity	HBV	hydrologic similarity based on FDC, similarity of drainage area and spatial proximity	2,3,6	FDC similarity produces better runoff simulation compared to the other three methods
Makungo et al. 2010	A quaternary catchment in Nzhelele River Catchment in South Africa (92Km ²)	Arid	Generating natural streamflow in ungauged catchment	Mike 11 NAM and AWBM (Australian Water Balance Model)	Modified nearest neighbour regionalization approach	2	The approach can be applied in near-real time modeling. NSE of verification period Mike 11 NAM : 0.74, AWBM : 0.77 .
Samaniego et al. 2010	38 German basins (70 to 4000 km ²)	Warm temperate	Using copula-based dissimilarity measures for daily streamflow simulation in ungauged basins	A conceptualized hydrologic model	Dissimilarity measures that are estimated from pair wise empirical copula densities of runoff	3	The achieved reasonable results NSE=0.76-0.86
Samuel et al. 2011	The main watersheds across the Province of Ontario, (85-100000 km ²)	Snowy	Estimation of daily flows in ungauged basins	MAC-HBV (McMaster-HBV)	Spatial proximity (i.e., kriging, inverse distance weighted (IDW) , spatial averaging), physical similarity, and regression-based approaches	2,3,5	An approach coupling the spatial proximity (IDW) method and the physical similarity produce better model performances than the remaining three
Bulygina et al. 2011	2 subcatchments in Wales (10.55km ² and 8.7 km ²)	Warm temperate	Using a formal Bayesian approach to estimate daily runoff	PDM model (Moore, 2007)	A probabilistic method (based on base flow index (BFI) and CN)	5	Nash–Sutcliffe efficiency ranging from 0.70 to 0.81 -both CN and HOST are potentially valuable sources of information for hydrological modeling of ungauged catchments within a stochastic modeling framework

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Table 3. Continued.

References	Study area	Climate	Temporal scale and key objectives	Hydrological model	Regionalization methods	Class	Major findings
Hundecha et al. 2008	101 sub watersheds of Rhine basin, German (400 - 2100 km ²)	Warm temperate	Estimation of model parameters from watershed attributes to estimate daily streamflow	Swedish Meteorological and Hydrological Institute(SMHI) and the HBV model	Kriging based on identifying the spatial structures of the parameters within a space defined using watershed physiographic and climatic attributes	2,3	NSE : 0.86 , slight improvement of performance compared to linear relationship between the model parameters and the watershed descriptors
Kim and Kaluarachchi 2008	Upper Blue Nile River Basin of Ethiopia (176,000 km ²)	Warm temperate and arid	Parameter estimation and monthly runoff regionalization in ungauged basins	A two-layer monthly water balance model with six parameters	Global mean, , multiple regression (regular , regional calibration aggregated calibration, and volume fraction calibration)	1,5	Regional calibration performed better in simulating the runoff of lumped basins than multiple regression (Nash–Sutcliffe efficiency=0.61 & 81)
Oudin et al. 2008	913 catchments; French catchments (10-9390 Km ²)	Warm temperate	Estimation of daily continuous flow in ungauged basins	GR4J and TOPMO(inspired by TOPMODEL)	Spatial proximity, physical similarity and multiple regression	2,3,5	Spatial proximity provides the best regionalization results- NSE=0.7-0.8
Jin et al. 2009	13 sub-basins in south-china (100-1000 km ²)	Warm temperate	Simulation of daily stream flow in ungauged basins	HBV	Proximity approach and global mean	1,2	NSE for global mean method is 0.74 and for second method is 0.72
Li et al. 2009	210 catchments in south-east Australia (area is not mentioned)	Arid to warm temperate	Prediction of daily runoff in ungauged catchments	Three versions of Xinanjiang rainfall–runoff model	The spatial proximity method and the physical similarity method	2,3	In terms of Nash–Sutcliffe Efficiency two method for the 3 versions give almost similar result (0.20-0.70)
Post 2009	24 catchments in the dry tropics of Australia (68–130.146km ²).	Arid	Estimation of daily streamflow in ungauged basins	IHACRES	Linear Regression	5	The predicted values of mean annual streamflow are within 20% of the observations (NSE around 0.7)
Reichl et al. 2009	184 Australian catchments (Area Kurtosis=3.214)	Arid to Warm temperate	Optimization of a similarity measure to produce the best monthly streamflow predictions	SIMHYD	Ensemble Techniques and Model Averaging model based on catchments similarities	3	The method is inferior to local calibration but it is superior to regression and spatial proximity (NSE=0.7-0.8)
Zhang and Chiew 2009	210 catchments in southeast Australia (51-2000Km ²)	Warm temperate	Daily runoff prediction in ungauged catchments	Xinanjiang and SIMHYD	Spatial proximity ,physical similarity, integrated similarity approach(combines the spatial proximity and physical similarity)	2,3	The spatial proximity approach performs slightly better than the physical similarity and the integrated similarity approach performs only very marginally better than the spatial proximity approach.

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Table 4. Studies on hydrologic model-independent streamflow regionalization methods (1990-2011)

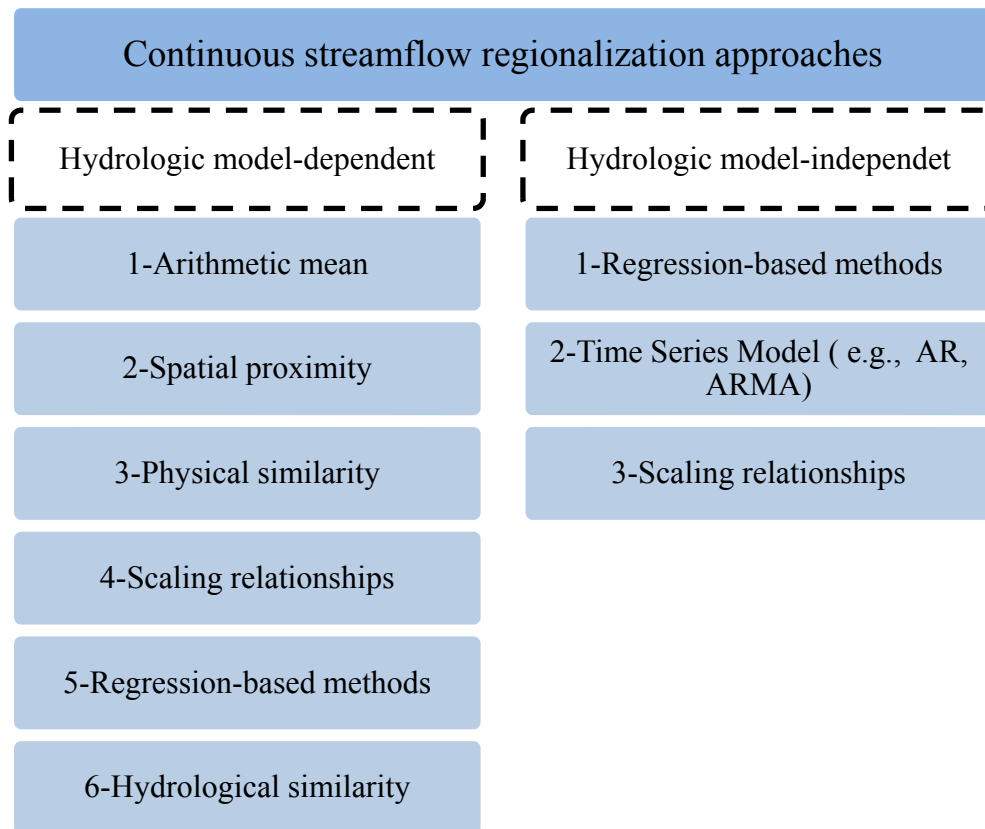
References	Study area	Climate	Temporal Scale and key objectives	Regionalization methods	Class	Major findings
Chiang et al. 2002	94 candidate stations in Alabama, USA (~37 to 30,810 mi ²)	Warm temperate	Using watershed characteristics to synthesize monthly streamflow hydrographs	Integrated Time series model and Multiple regression analysis (MRA) to estimate TSM parameters using catchment attributes	1,2	Integrated Time series model outperform simple regression analysis method
Goswami et al. 2007	12 basins in France (32.1 - 371 km ²)	Warm temperate	Estimation of daily continuous flow in ungauged basins	Parametric and non-parametric simple linear model, perturbation model, linearly varying gain factor, ANN	1	The non-linear ANN is best in nine out of the 12 catchments in calibration
Yadav et al. 2007	30 small to medium sized watersheds in the UK (50–1100 km ²)	Warm temperate	Ensemble predictions of daily flow in ungauged basins in an uncertainty framework	Stepwise linear regression between 39 dynamic response characteristics & 13 physical watershed characteristics	1	This approach is not impacted by problems of parameter calibration or model structural uncertainty and could guide an improved approach to watershed classification
Buttle and Eimers 2009	22 small basins (3.4–190.5 ha) on the Precambrian Shield in south-central Ontario, Canada	Snowy	Explain inter-basin variations in streamflow metrics in terms of basin scale and physiographic and predict annual maximum and mean daily stream flow	Scaling relationships based on basin size and physiographic properties	3	Annual maximum and mean daily stream flow can be well-predicted using simple scaling relationships, (NSE=0.21-0.63)
Isik and Singh (2008)	26 river basins in Turkey (~700000km ² in total)	Warm temperate and Snowy parts	Regionalization of monthly streamflow in Watersheds of Turkey	FDCs model after basin classification using k-means partitioning method	1	Monthly discharge data with correlation coefficients between 95 and 100% for 85.7% of the gauging stations
Mohamoud 2008	29 catchments n Mid-Atlantic Region, US (23-4250 Km ²)	Warm temperate	Prediction of daily flow duration curves (FDCs) and streamflow for ungauged catchments	Regional FDC & drainage area ratio methods and Step-wise multiple regression analysis using catchment descriptors to find the parameters	1	FDC-based method shows great promise for predicting streamflow in ungauged basins.NSE=0.60-0.97
Besaw et al. 2010	The Winooski River basin, northwestern Vermont, USA (2700 km ²)	Snowy	Streamflow forecast (hourly and daily) in ungauged basins	GRNN and CPN with recurrent feedback loops, linear regression and time series autoregressive moving average	1,2	Both ANNs better captured the climate-flow relationships when trained on hourly data reflecting the basin-scale characteristics-CPN, GRNN and MLR provide the most accurate and unbiased estimates of stream flow
Lima and Lall 2010	40 hydropower sites in Brazil (2588 to 823,555 km ²)	Equatorial	Modeling of non-stationary monthly streamflow series and annual maximum flood series	Hierarchical Bayesian models : the Bayesian estimation of the scaling relationship with drainage area for annual maximum flows and monthly streamflow accounting for nonstationary and seasonal variability	3	Statistics of the streamflow time series in particular moments scale with physical properties of the drainage basin (The correlation between simulated and observed flow=0.63-0.97)
Parada & Liang 2010	North Fork Cache Creek watershed (510 km ²) and Bluestone River watershed (1023 km ²), USA	Warm temperate	Inference of daily streamflow for ungauged basins combining concepts from both kernel methods and data assimilation	Kernel-based inference method	1	The inference method can estimate streamflow for ungauged basins with unknown and nonlinear dynamics. RMSE=0.37 & 0.21-Bias=0.23 & 0.02

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List of Figures

Fig.1.Schematic of two main classes and subdivisions of continuous streamflow regionalization methods

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